

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 8

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Apache Benchmarking, Apache Security and CMS Installation

1. Run your Linux Virtual Machine in VirtualBox and install ApacheBench (ab) tool in your Linux machine using `sudo apt-get install apache2-utils`.

```
> sudo apt-get install apache2-utils
```

```
> y
```

OR

```
> sudo apt install lamp-server^
```

```
osboxes@osboxes:~$ sudo apt-get install apache2-utils
[sudo] password for osboxes:
Reading package lists... Done
Building dependency tree
Reading state information... Done
apache2-utils is already the newest version (2.4.41-4ubuntu3.21).
apache2-utils set to manually installed.
The following packages were automatically installed and are no longer required:
  libssh2-1 mc-data
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 156 not upgraded.
1 not fully installed or removed.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Setting up initramfs-tools (0.136ubuntu6.7) ...
update-initramfs: deferring update (trigger activated)
```

Note: If you install LAMP server using `sudo apt install lamp-server^` the `apache2-utils` also install by default

a) Run ab tool to perform benchmarking for <http://www.mmu.edu.my/> with 200 requests to perform and 20 multiple requests at a time. Identify **time taken for test** and **Web Server used**.

```
> ab -n 200 -c 20 http://www.mmu.edu.my/
```

```
osboxes@osboxes:~$ ab -n 200 -c 20 http://www.mmu.edu.my/
This is ApacheBench, Version 2.3 <$Revision: 1843412 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking www.mmu.edu.my (be patient)
Completed 100 requests
Completed 200 requests
Finished 200 requests

Server Software:
Server Hostname:      www.mmu.edu.my
Server Port:          80

Document Path:        /
Document Length:      862 bytes

Concurrency Level:     20
Time taken for tests:   1.278 seconds
Complete requests:     200
Failed requests:        120
  (Connect: 0, Receive: 0, Length: 120, Exceptions: 0)
Non-2xx responses:     200
Total transferred:     274576 bytes
HTML transferred:     172488 bytes
Requests per second:   156.55 [#/sec] (mean)
Time per request:      127.756 [ms] (mean)
Time per request:      6.388 [ms] (mean, across all concurrent requests)
Transfer rate:         209.89 [Kbytes/sec] received

Connection Times (ms)
              min      mean[+/-sd] median    max
Connect:     28       47   11.2      46     102
Processing:   39       69   16.7      70     130
Waiting:     38       69   16.8      69     130
Total:       72      116   21.7     113     192

Percentage of the requests served within a certain time (ms)
 50%    113
 66%    124
 75%    128
 80%    131
 90%    142
 95%    159
 98%    176
 99%    191
100%    192 (longest request)
```

Time taken for tests: 1.278 seconds

Web server used (Server software): Apache

> By using `curl -I http://mmu.edu.my`, it shows HTTP header directly, including the Server Header.

```
osboxes@osboxes:~$ curl -I http://mmu.edu.my
HTTP/1.1 301 Moved Permanently
Date: Thu, 02 Jan 2025 02:48:05 GMT
Server: Apache
Location: https://www.mmu.edu.my/
Content-Type: text/html; charset=iso-8859-1
Set-Cookie: visid_incap_2990630=HqbgceRYSQy51Hx1GUFdath6dWcAAAAQUIPAAAAACiX1NXgmnbcKU7rl0ocp0l; expires=Thu, 01 Jan 2026 16:20:56 GMT; HttpOnly; path=/; Domain=.mmu.edu.my
Set-Cookie: incap_ses_1132_2990630=bGfjSRcoTXEjeurnM6y1D9H6dWcAAAAA+uLu33IIszochgQOSIoQ8wQ==; path=/; Domain=.mmu.edu.my
X-CDN: Imperva
X-Iinfo: 12-101951178-101951188 NNNN CT(6 -1 0) RT(1735785169514 20) q(0 0 0 0) r(0 0) U6
```

b) Perform similar test as above to <http://www.mudah.my/>. Identify time taken for test and Web Server used. Give comment on both results.

```
> ab -n 200 -c 20 http://www.mudah.my/
```

```
osboxes@osboxes:~$ ab -n 200 -c 20 http://www.mudah.my/
This is ApacheBench, Version 2.3 <$Revision: 1843412 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/

Benchmarking www.mudah.my (be patient)
Completed 100 requests
Completed 200 requests
Finished 200 requests


Server Software:        cloudflare
Server Hostname:        www.mudah.my
Server Port:            80

Document Path:          /
Document Length:        167 bytes

Concurrency Level:      20
Time taken for tests:    2.126 seconds
Complete requests:      200
Failed requests:         0
Non-2xx responses:      200
Total transferred:      169238 bytes
HTML transferred:       33400 bytes
Requests per second:    94.06 [#/sec] (mean)
Time per request:       212.624 [ms] (mean)
Time per request:       10.631 [ms] (mean, across all concurrent requests)
Transfer rate:          77.73 [Kbytes/sec] received

Connection Times (ms)
  min   mean[+/-sd] median   max
Connect:    34    76  28.3    66   167
Processing:  43   122  59.8   115   555
Waiting:    43   119  59.1   111   554
Total:      101   197  63.4   188   619

Percentage of the requests served within a certain time (ms)
 50%    188
 66%    212
 75%    226
 80%    238
 90%    267
 95%    289
 98%    342
 99%    539
100%    619 (longest request)
```

Time taken for test: 2.126 seconds

Web server used (Server Software): Cloudflare

> By using curl -I <http://mudah.my>, we could also see the info about the website.

```
osboxes@osboxes:~$ curl -I http://mudah.my
HTTP/1.1 301 Moved Permanently
Date: Thu, 02 Jan 2025 02:34:37 GMT
Content-Type: text/html
Content-Length: 167
Connection: keep-alive
Cache-Control: max-age=3600
Expires: Thu, 02 Jan 2025 03:34:37 GMT
Location: https://www.mudah.my/
Report-To: {"endpoints":[{"url":"https://a.nel.cloudflare.com/report/v4?s=IezGZftmoXIcJHr
eFYEM7t3KeDg7cTsYyMSfNhkFgtwvD4h0%2F8JzbmsQp04G6fU08ywlB7MjGQgzmtHrD6CadCMsDmSdMHBtbn2j50deTF
GZVDj8Lgt19VS6"}],"group":"cf-nel","max_age":604800}
NEL: {"success_fraction":0.01,"report_to":"cf-nel","max_age":604800}
Server: cloudflare
CF-RAY: 8fb759e23a1ed431-KUL
alt-svc: h3=":443"; ma=86400
```

2. When a directory is requested, Apache may be configured to send a particular file within that directory automatically. This is configured with the `DirectoryIndex` directive. To prevent directory listings (for security purposes, for example), you should remove the `Indexes` keyword from every `Options` directive in your configuration file.

a) Identify if your `/var/www/html` enable directory listing.

> `curl -I localhost`

```
osboxes@osboxes:~$ curl -I localhost
HTTP/1.1 200 OK
Date: Thu, 02 Jan 2025 02:44:25 GMT
Server: Apache/2.4.41 (Ubuntu)
Last-Modified: Thu, 26 Dec 2024 02:32:16 GMT
ETag: "2aa6-62a232796e0f5"
Accept-Ranges: bytes
Content-Length: 10918
Vary: Accept-Encoding
Content-Type: text/html
```

b) Disable the directory listing in `/var/www/html`.

> `sudo nano /etc/apache2/apache2.conf`

```
# access here, or in any related virtual host.
<Directory />
    Options FollowSymLinks
    AllowOverride None
    Require all denied
</Directory>

<Directory /usr/share>
    AllowOverride None
    Require all granted
</Directory>

<Directory /var/www/>
    Options Indexes FollowSymLinks
    AllowOverride None
    Require all granted
    DirectoryIndex pmwiki.php index.php index.html
</Directory>

<Directory /var/www/html>
    Options -Indexes
</Directory>
```

> `sudo systemctl restart apache2.service`

```
osboxes@osboxes:/var/www/html$ curl -I localhost
HTTP/1.1 200 OK
Date: Thu, 02 Jan 2025 02:53:06 GMT
Server: Apache/2.4.41 (Ubuntu)
Last-Modified: Thu, 26 Dec 2024 02:32:16 GMT
ETag: "2aa6-62a232796e0f5"
Accept-Ranges: bytes
Content-Length: 10918
Vary: Accept-Encoding
Content-Type: text/html
```

3. Apache will display the Apache version and OS identity on page error. Edit your Apache configuration `/etc/apache/apache2.conf` and insert two directives at the end of the configuration file to hide the Apache version and OS identity:

- `ServerSignature Off`
- `ServerTokens Prod`

> `sudo nano /etc/apache2/apache2.conf`

```
ServerSignature Off
ServerTokens prod
```

> `sudo systemctl restart apache2.service`

> `curl -I localhost`

```
osboxes@osboxes:/var/www/html$ curl -I localhost
HTTP/1.1 200 OK
Date: Thu, 02 Jan 2025 02:57:17 GMT
Server: Apache
Last-Modified: Thu, 26 Dec 2024 02:32:16 GMT
ETag: "2aa6-62a232796e0f5"
Accept-Ranges: bytes
Content-Length: 10918
Vary: Accept-Encoding
Content-Type: text/html
```

4. WordPress is one of the best CMS around, and it is used by 63.6% of website in the Internet.

a) Download and install WordPress in Apache Server of your Linux Virtual Machine client.

- Download the WordPress to Server root /var/www/html. In Server root give command: `sudo wget https://wordpress.org/latest.zip`

> `cd /var/www/html`

> `sudo wget https://wordpress.org/latest.zip`

```
osboxes@osboxes:/var/www/html$ sudo wget https://wordpress.org/latest.zip
--2025-01-02 02:59:44-- https://wordpress.org/latest.zip
Resolving wordpress.org (wordpress.org)... 198.143.164.252
Connecting to wordpress.org (wordpress.org)|198.143.164.252|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 28585184 (27M) [application/zip]
Saving to: 'latest.zip'

latest.zip           53%[=====>          ] 14.56M   132KB/s   eta 27s
```

> `ls`

```
osboxes@osboxes:/var/www/html$ ls
index.html  pmwiki          sysadmin.html      systemadmin.html.save.1
info.php    pmwiki-2.3.38   systemadmin.html
latest.zip  pmwiki-latest.zip systemadmin.html.save
```

- Unzip the WordPress, unzip latest.zip

> `sudo unzip latest.zip`

```
osboxes@osboxes:/var/www/html$ sudo unzip latest.zip
Archive: latest.zip
  creating: wordpress/
  inflating: wordpress/xmlrpc.php
  inflating: wordpress/wp-blog-header.php
  inflating: wordpress/readme.html
  inflating: wordpress/wp-signup.php
  inflating: wordpress/index.php
  inflating: wordpress/wp-cron.php
  inflating: wordpress/wp-config-sample.php
  inflating: wordpress/wp-login.php
  inflating: wordpress/wp-settings.php
  inflating: wordpress/license.txt
  creating: wordpress/wp-content/
  creating: wordpress/wp-content/themes/
  creating: wordpress/wp-content/themes/twentytwentythree/
  inflating: wordpress/wp-content/themes/twentytwentythree/theme.json
```

> To verify wordpress directory is there: `ls -l`

```
osboxes@osboxes:/var/www/html$ ls -l
total 28616
-rw-r--r-- 1 root    root      10918 Dec 26 02:32 index.html
-rw-r--r-- 1 root    root         21 Dec 26 02:48 info.php
-rw-r--r-- 1 root    root  28585184 Nov 21 14:08 latest.zip
drwxr-xr-x 7 www-data www-data   4096 Dec 27 15:45 pmwiki
drwxr-xr-x 8 root     root       4096 Nov  3 11:03 pmwiki-2.3.38
-rw-r--r-- 1 root    root    671561 Nov  3 11:04 pmwiki-latest.zip
-rw-r--r-- 1 root    root       222 Dec 26 03:15 sysadmin.html
-rw-r--r-- 1 root    root       177 Dec 26 03:11 systemadmin.html
-rw-r--r-- 1 root    root       170 Dec 26 06:17 systemadmin.html.save
-rw-r--r-- 1 root    root       682 Dec 26 06:17 systemadmin.html.save.1
drwxr-xr-x 5 root     root       4096 Nov 21 14:07 wordpress
```

> `sudo chown -R www-data:www-data /var/www/html/`

> `sudo chmod -R 755 /var/www/html`

```
osboxes@osboxes:/var/www/html$ sudo chown -R www-data:www-data /var/www/html
osboxes@osboxes:/var/www/html$ sudo chmod -R 755 /var/www/html
```

◦ Using Linux terminal, login to MySQL Server using root. Create database and user for WordPress. Give ALL permissions to database user to access the database.

- Database: wordpress
- Database User: user
- Database Password: user123

> sudo apt update

> sudo apt install mysql-server

> sudo mysql -u root -p

```
osboxes@osboxes:~$ sudo mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.40-0ubuntu0.20.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE wordpress;
Query OK, 1 row affected (0.01 sec)

mysql> CREATE USER 'user'@'localhost' IDENTIFIED BY 'user123';
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON wordpress.* TO 'user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.01 sec)

mysql> EXIT;
Bye

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
| wordpress |
+-----+
5 rows in set (0.01 sec)

mysql> SHOW GRANTS FOR 'user'@'localhost';
+-----+
| Grants for user@localhost |
+-----+
| GRANT USAGE ON *.* TO `user`@`localhost` |
| GRANT ALL PRIVILEGES ON `wordpress`.* TO `user`@`localhost` |
+-----+
2 rows in set (0.00 sec)
```

◦ Perform the web-based installation, run the web browser. Point your browser at <http://server-ip-address/wordpress> and enter database information above to complete the installation.

> cd /var/www/html

> sudo mv wordpress/* .

> ls

```
osboxes@osboxes: /var/www/html$ ls
index.html      systemadmin.html      wp-cron.php
index.php       systemadmin.html.save wp-includes
info.php        systemadmin.html.save.1 wp-links-opml.php
latest.zip      wordpress             wp-load.php
license.txt     wp-activate.php       wp-login.php
pmwiki          wp-admin              wp-mail.php
pmwiki-2.3.38  wp-blog-header.php    wp-settings.php
pmwiki-latest.zip wp-comments-post.php  wp-signup.php
readme.html     wp-config-sample.php  wp-trackback.php
sysadmin.html   wp-content            xmlrpc.php
```

> sudo cp wp-config-sample.php wp-config.php

> sudo nano wp-config.php

```
GNU nano 4.8 wp-config.php
* @link https://developer.wordpress.org/advanced-administrati
*
* @package WordPress
*/

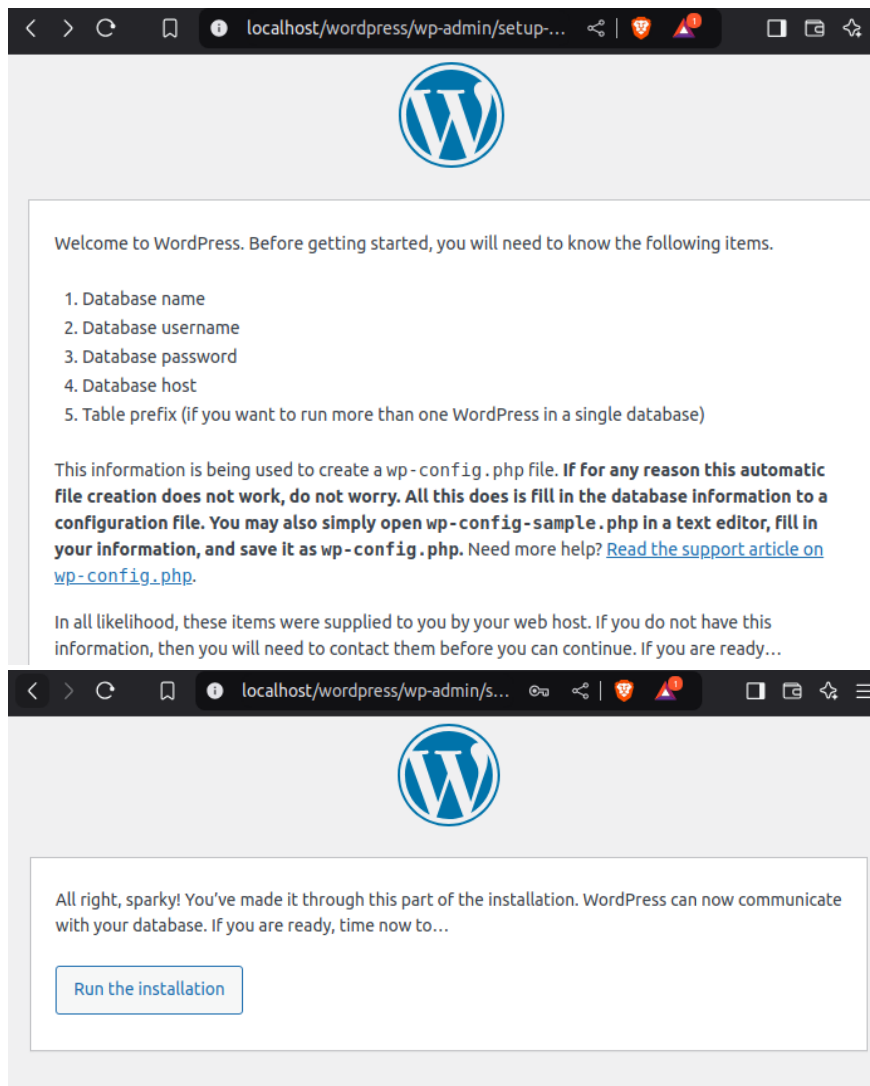
// ** Database settings - You can get this info from your web
/** The name of the database for WordPress */
define( 'DB_NAME', 'wordpress' );

/** Database username */
define( 'DB_USER', 'user' );

/** Database password */
define( 'DB_PASSWORD', 'user123' );

/** Database hostname */
define( 'DB_HOST', 'localhost' );
```

b) Access, display and capture your WordPress default page using any web browser.



localhost/wordpress/wp-admin/i...

Information needed

Please provide the following information. Do not worry, you can always change these settings later.

Site Title

Username

Username can have only alphanumeric characters, spaces, underscores, hyphens, periods, and the @ symbol.

Password

Strong

localhost/wordpress/wp-admin/i...



Success!

WordPress has been installed. Thank you, and enjoy!

Username

ae1207

Password

Your chosen password.

[Log In](#)

localhost/wordpress/wp-admin/

Screen Options ▾

Help ▾

Dashboard

×

Welcome to WordPress!

[Learn more about the 6.7.1 version.](#)



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